

Evaluating Road Safety: A Predictive Algorithm for Accident Risk Levels



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INTRODUCTION

Objective: Integrate a predictive machine learning model into Google Maps to forecast road risk levels.

Implementation: Demonstrated through a Streamlit app and visualized integration plans using Figma.

Purpose: To equip drivers with real-time data, enhancing decision-making and aiming to reduce road accidents in Breda.

Problem statement

Global Crisis: Every year, road traffic accidents claim over 1.2 million lives globally, highlighting a critical need for enhanced road safety measures.

Rising Trends: Recent statistics from the Netherlands show a worrying 20% rise in road-related fatalities and serious injuries, indicating a growing problem.

Technological Gap: Current navigation technologies do not adequately predict or mitigate risks, necessitating advanced solutions for accurate road safety assessments.





Data Description

For our AI predictive algorithm for accident risk levels, we used two data sources:

- ANWB Traffic Incidents Data, which contains important traffic related features like the duration of a incident or the longitude and latitude where the incident took place.
- Breda Accident Data from 2017 to 2023, which contains useful information about the road condition and the weather related to accidents in Breda.

The aforementioned data enables a comprehensive risk assessment by enabling us to work with good, well structured data.



Data Collection

Methodology

For the data collection step, we joined the two datasets at our disposal, leaving us with the following features:

- From the Breda Accidents Data:
 - Accident severity - target feature
 - First Mode of Transport
 - Area Type
 - Light condition
 - Road Location
 - Road condition
 - Road surface
 - Road situation
 - Speed limit
 - weather
 - accidents
- From the ANWB Traffic Incidents Data:
 - Average longitude
 - Average latitude



Methodology

Data Preprocessing

For the Preprocessing step, we used the following methods to prepare data:

- **Missing Values Analysis.** We conducted the missing value analysis by checking the data for missing values and then deciding whether the missing values bring any value to the project or if we should eliminate them from the data.
- **Outliers Analysis.** We conducted outliers analysis by looking at each independent feature and establishing the frequency analysis, since most of the features use were categorical.
- **Label Encoding.** We used label encoding for each categorical feature so that we can use the data to develop a model.
- **Standardization.** We standardized the data so that it can be more interpretable for the model.
- **Random Oversampling.** We used random oversampling to balance our target feature, since the three classes that were used for predicting were badly skewed.

Risk Level



High Risk Classification

EU AI Act identifies significant safety implications and data sensitivity.



Key Concerns

Safety Impact: Errors in predictions could lead to hazardous situations.

Data Privacy: Compliance with GDPR is critical due to extensive data usage.

Decision-Making Influence: Accuracy is vital as it affects driver and authority decisions.



Risk Management Strategies

Continuous Monitoring: Ensure system accuracy and reliability through regular updates.

Legal Compliance: Adhere to data protection laws and AI safety regulations.

User Transparency: Provide clear guidelines on system use and limitations to maintain user awareness.

Legal Requirements and Obligations

A

SEARCH THE Data Accuracy and Integrity (Article 10)

Requirement: Use accurate, relevant, and representative data.

Obligation: Regular validation and updating of data sources to prevent biases and ensure current, comprehensive data usage.

B

Documentation and Record-keeping (Article 12):

Requirement: Maintain detailed records of development processes, data handling, and testing methodologies.

Obligation: Ensure documentation is available for audits and compliance, promoting transparency and accountability.

C

Transparency and Information Provision (Article 13):

Requirement: Be transparent about functionalities, decision-making processes, and prediction methods.

Obligation: Provide users with clear, accessible explanations of data use and prediction limitations.

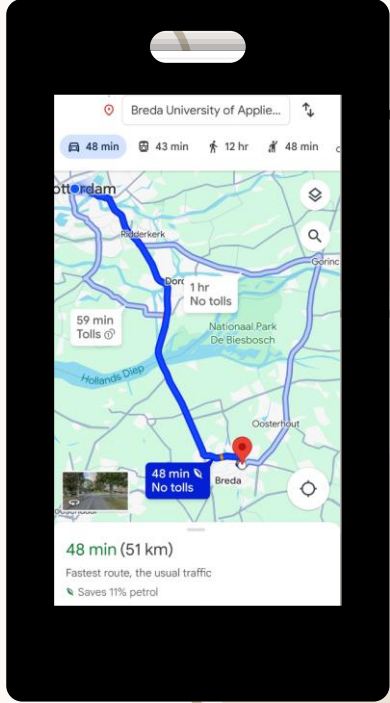
Streamlit App



Google Maps
Extension
Proposal

Figma Design

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Thanks!

Do you have any questions?

